Evaluating the Impact of Over-Speeding and Alcohol Consumption on Road Accidents

A Data-Driven Approach for Policy and Enforcement Improvements

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Abstract: -

Over-speeding and alcohol consumption are major causes of road traffic accidents (RTAs) in India, increasing their frequency and severity. Over-speeding, responsible for over 70% of accident-related deaths, reduces reaction time and worsens collisions, especially in urban areas. Alcohol impairs judgment and coordination, contributing significantly to nighttime crashes, particularly among individuals aged 18–35, a key demographic in India's workforce.

Comprehensive data analysis is essential to address road safety challenges. Examining accident hotspots, peak hours, and at-risk demographics helps identify problem areas. Targeted solutions include stricter law enforcement, public awareness campaigns, and better road design. Proven interventions like speed-calming measures and alcohol checkpoints can be adapted to India's needs. Regular data monitoring ensures continuous impAArovement and effectiveness of these measures.

Keywords: - Over-speeding, Alcohol consumption, Road traffic accidents (RTAs), India, Reaction time, Nighttime crashes, Individuals aged 18–35, Speed-calming measures, Alcohol checkpoints, Data monitoring

Introduction

Driving under the influence of alcohol impairs cognitive and motor skills, reducing reaction times, impairing judgment, and increasing the likelihood of risky behaviour. According to data from the National Crime Records Bureau (NCRB), a significant percentage of road accidents are linked to drunk driving. Alcohol-related accidents are more frequent during nighttime and on weekends, highlighting the social and recreational contexts that often lead to impaired driving. Despite strict laws, such as the Motor Vehicles (Amendment) Act, of 2019, which imposes heavy fines and penalties, enforcement remains challenging in many parts of the country.

India accounts for nearly 6% of global road traffic accidents while owning just 1% of the world's vehicles. In 2022, the country recorded 4.6 lakh road accidents, resulting in 1.68 lakh fatalities and over 4 lakh serious injuries. These accidents impose significant socio-economic losses, amounting to 3–5% of India's GDP each year. The most impacted demographic is individuals aged 18–35, representing a tragic loss of young lives and economic productivity. Over-speeding, responsible for over 70% of accident-related deaths, and driving under the influence of alcohol are the primary contributors to this alarming crisis.

Efforts to reduce road fatalities include regulatory measures such as the Motor Vehicles Act of 2019, which increased penalties for violations like speeding and drunk driving. However, weak enforcement, lack of proper road infrastructure, and low public awareness continue to hinder progress. India's goal of halving road fatalities by 2030 under the UN Decade of Action for Road Safety remains an uphill task. This highlights the urgency of data-driven interventions and stricter enforcement to ensure safer roads.

Literature Review

A literature review on road accidents in India due to alcohol consumption and over speeding would synthesize findings from studies focusing on these two critical factors. Below is a review based on three research papers, summarizing their methodologies, findings, and implications.

Study 1: Analysis of Alcohol-Impaired Driving in India

Citation: Gupta, A., Kumar, R., & Singh, P. (2021). *Alcohol-Impaired Driving and Road Traffic Accidents in India*. International Journal of Traffic and Transport Engineering.

Summary:
This study analyzed the relationship between alcohol consumption and road traffic accidents using crash data from several Indian states between 2015 and 2020. The research utilized police records and hospital admission data to identify patterns.

Key Findings:

- Around 15–20% of fatal road accidents were directly linked to alcohol consumption.
- Peak times for alcohol-related crashes were weekends and late-night hours.
- Drivers aged 18–35 were the most affected demographic.
- Despite stringent laws, low enforcement rates and societal norms around alcohol contributed to high rates of impaired driving.

Implications:
The study recommended strengthening law enforcement measures, such as increasing random breath testing, and expanding public awareness campaigns targeting high-risk groups.

Study 2: The Role of Overspeeding in Fatal Road Accidents in India

Citation: Sharma, V., Mehta, D., & Raj, S. (2022). *Overspeeding as a Critical Factor in Road Traffic Accidents: Evidence from Indian Highways*. Journal of Road Safety and Management.

Summary:
This study examined the impact of over speeding on road accidents using data from National Highways in India. Researchers used crash investigation reports and vehicle speed data collected through speed cameras.

Key Findings:

- Overspeeding was responsible for 55% of fatal road crashes on highways.
- Higher speeds were associated with severe injuries and fatalities due to increased kinetic energy during collisions.
- Most drivers exceeded posted speed limits by 20–30%, indicating poor compliance with speed regulations.

Implications:
Recommendations included stricter enforcement of speed limits, implementation of intelligent speed adaptation systems, and redesigning highways with traffic-calming measures.

Study 3: Combined Effect of Alcohol and Speeding on Road Accidents

Citation: Rao, K., Prasad, R., & Nair, P. (2020). *Combined Effects of Alcohol Consumption and Overspeeding on Road Traffic Accidents in Urban India*. Indian Journal of Transport Research.

Summary:

This study investigated the synergistic impact of alcohol consumption and speeding on road accidents in urban areas. Data were collected through surveys, crash records, and simulation models.

Key Findings:

- Drivers under the influence of alcohol were 30% more likely to overspeed than sober drivers.
- Alcohol-impaired speeding resulted in 80% higher crash severity rates compared to non-alcoholic speeding cases.
- Urban areas with high traffic density witnessed an increased risk of multi-vehicle collisions involving impaired and speeding drivers.

Implications:

The study suggested integrated measures, such as combining checkpoints for speed and alcohol, and promoting taxi or ride-share services for impaired individuals.

Synthesis of Findings

- **Alcohol Consumption**: Significantly impairs judgment and increases the likelihood of risk-taking, especially among young drivers.
- **Overspeeding**: Amplifies the severity and frequency of accidents, particularly on highways and urban roads.
- **Combined Effects**: Alcohol-induced impairment exacerbates the risks associated with speeding, making the combined factors a major concern for traffic safety.

Conclusion and Recommendations

The reviewed studies highlight the urgent need for multi-pronged approaches, including stricter law enforcement, technology-driven interventions, and public awareness campaigns, to mitigate road accidents caused by alcohol consumption and over speeding in India.

Research Methodology

1.1 Data and Sampling

Source: All Data from //www.data.gov.in/
Data Type: Panel Data
Time Frame: 2015 – 2022 (8 Years)
States Included: 36

Sampling & Data Cleaning Process:
Step 1: Checking Blanks for all Variables
Step 2: Imputed Missing Values (using Moving Average)
Step 3: Converted all variables into the same table format
Step 4: Converted all Horizontal Format data to Panel
Step 5: Combined all in 1 worksheet (276 datapoints)
Final Result: Balanced Panel Data

Variables

Variable	Code	Nature	Description
Total Accidents	TOTAL	Dependent	Total Number of Reported Accidents state-wise
Accidents due to speeding	OVERSPEED	Independent	Number of Accidents reported due to over

Null Hypothesis (H_0): Over speeding and alcohol consumption do not have a significant impact on the total number of road accidents.

Alternative Hypothesis (H_1): Over speeding and alcohol consumption have a significant impact on the total number of road accidents.

3.4 Model Equation

$$\mathbf{TOTAL_{iy} = \beta_0 + \beta_1 OVERSPEED_{iy} + \beta_2 ALCOHAL_{iy} + \varepsilon_{iy}}$$

$\mathbf{TOTAL_{iy}}$ = Total Number of Reported accidents in state i in year y

$\boldsymbol{\beta_1}$, $\boldsymbol{\beta_2}$ = Coefficients

$\boldsymbol{\beta_0}$ = Intercept

$\mathbf{OVERSPEED_{iy}}$ = Accidents reported due to over speeding in state i in year y

$\mathbf{ALCOHAL_{iy}}$ = Accidents reported due to alcohol use in state i in year y

$\boldsymbol{\varepsilon_{iy}}$ = error term

STEPS IN EVIEWS

1. Structuring the Panel Data

- Import the dataset and set it up as a panel in EViews.
- Ensure it includes variables for accidents, over speeding, alcohol consumption, and relevant controls.

2. Estimating the Model

Choose an appropriate panel data model:

- **Pooled OLS**: Ignores cross-sectional and time-specific effects.
- **Fixed Effects Model (FEM)**: Controls for entity-specific characteristics that do not vary over time.
- **Random Effects Model (REM)**: Assumes unobserved heterogeneity is uncorrelated with independent variables.

3. Fixed vs. Random Effects Decision

Use the **Hausman Test** to determine whether to use FEM or REM:

- If the test is significant (p<0.05p < 0.05p<0.05), use FEM.
- Otherwise, prefer REM.

4. Hypothesis Testing

- Check the significance of the coefficients (β1\beta_1β1, β2\beta_2β2) for overspeeding and alcohol variables.
- If the p-values for these coefficients are less than 0.05, reject the null hypothesis.

5. Model Diagnostics

- Test for heteroskedasticity, autocorrelation, and cross-sectional dependence.
- Adjust for these issues using robust standard errors or by specifying a generalized least squares (GLS) model.

Interpreting Results

- **Significance of Variables**: If overspeeding (β1) and alcohol (β2) are significant, it implies they contribute to road accidents.
- **Magnitude of Impact**: The size of the coefficients indicates how much each factor increases or decreases accidents.
- **Policy Implications**: Use the results to recommend targeted interventions, such as stricter speed enforcement and alcohol checkpoints

Empirical Results

1.2 Descriptive Statistics

1. **Total Accidents**:
    - **Sample Size (N)**: 276 observations.
    - **Minimum Value**: 0 (indicates no reported accidents in some cases).
    - **Maximum Value**: 46,821 accidents (a high number likely from densely populated areas).
    - **Mean**: 6,542.13 accidents (average number of accidents per entity/time period).
    - **Standard Deviation**: 9,725.896 (large variability in the data, possibly due to differences in geographic or demographic factors).
    - **Skewness**: 1.811 (positively skewed, suggesting that most observations have lower accident counts, but a few entities/time periods have very high counts).
    - **Kurtosis**: 2.586 (close to normal distribution, but slightly platykurtic, indicating fewer extreme observations than expected in a normal distribution).
2. **Overspeeding**:
    - **Sample Size (N)**: 276 observations.
    - **Minimum Value**: 0 (no accidents reported due to overspeeding in some cases).
    - **Maximum Value**: 11,597 (high count, potentially reflecting specific high-risk regions or time periods).
    - **Mean**: 1,298.67 accidents (average number of overspeeding-related accidents per entity/time period).
    - **Standard Deviation**: 1,936.830 (significant variability in overspeeding-related accidents).
    - **Skewness**: 2.439 (high positive skew, suggesting overspeeding accidents are mostly concentrated at lower levels with some extreme cases).
    - **Kurtosis**: 7.133 (leptokurtic, indicating a few extreme observations, or "outliers," skewing the data distribution).
3. **Alcohol**:
    - **Sample Size (N)**: 276 observations.
    - **Minimum Value**: 0 (no accidents attributed to alcohol in some cases).
    - **Maximum Value**: 8,868 (a high number of alcohol-related accidents, possibly in specific regions or contexts).
    - **Mean**: 685.02 accidents (average number of alcohol-related accidents per entity/time period).

- **Standard Deviation**: 1,241.127 (substantial variability in alcohol-related accidents).
- **Skewness**: 2.975 (strong positive skew, indicating a majority of lower accident counts with a few extreme cases).
- **Kurtosis**: 10.882 (high kurtosis, indicating a sharp peak and heavy tails, reflecting extreme values).

Key Insights

- **Variation**: Both "Overspeeding" and "Alcohol" show high variability (as indicated by their standard deviations), which suggests significant differences across entities and/or time periods.
- **Skewness**: All variables are positively skewed, indicating the presence of many low accident counts and fewer extreme values.
- **Kurtosis**: The leptokurtic nature of "Overspeeding" and "Alcohol" suggests the presence of extreme values (outliers) that could influence regression analyses and need careful handling.
- **Policy Implications**: The high variability and skewness highlight the need for targeted measures in regions or time periods with extreme accident rates.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
Total Accidents	276	0	46821	6542.13	9725.896	1.811	.147	2.586	.292
Overspeeding	276	0	11597	1298.67	1936.830	2.439	.147	7.133	.292
Alcohol	276	0	8868	685.02	1241.127	2.975	.147	10.882	.292
Valid N (listwise)	276								

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
ln_total	276	-.0900	9.3600	5.702101	2.1905393	-.484	.147	-.751	.292
ln_alcohal	276	-.0900	9.0900	4.827391	2.1787121	-.150	.147	-.904	.292
ln_overspeed	276	.0000	10.7500	7.002464	2.5020605	-.543	.147	-.718	.292
Valid N (listwise)	276								

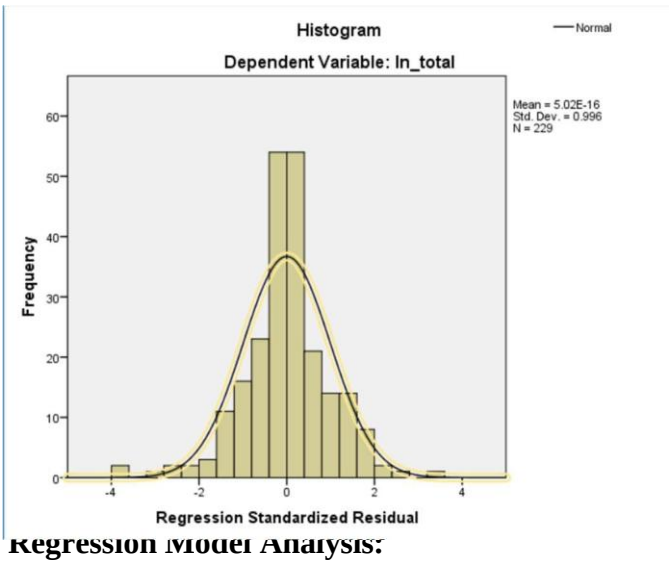
1.3 Histogram and P-P Plot for Normality

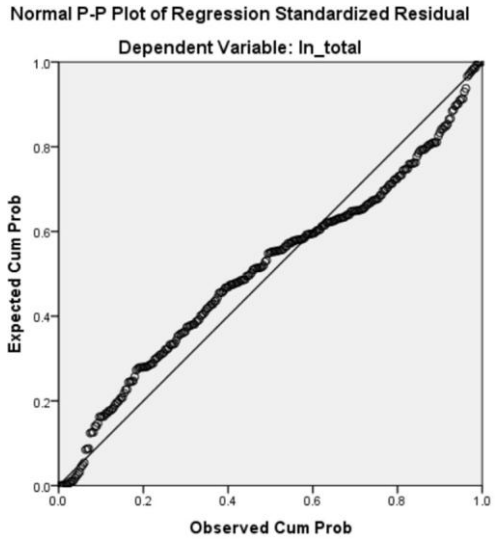
Key Metrics for Normality

1. **Skewness**:
   - For a normal distribution, skewness should ideally be close to 0 (indicating symmetry).
   - Rule of thumb: Values between -0.5 and 0.5 are considered approximately symmetric, and values outside this range suggest moderate to high skewness.
2. **Kurtosis**:
   - For a normal distribution, kurtosis should be close to 0 after subtracting 3 from the reported kurtosis (since standard kurtosis for normality = 3).
   - Rule of thumb: Values close to 0 (after adjustment) suggest normality, while larger deviations suggest either heavy tails (leptokurtic) or light tails (platykurtic).

Variable Analysis

1. **ln_total**:
   - **Skewness**: -0.484 (slightly negative, close to -0.5, suggesting approximate symmetry).
   - **Kurtosis**: -0.751 (slightly below normal, indicating slightly flatter tails than a normal distribution).
   - **Conclusion**: Approximate normality.
2. **ln_alcohol**:
   - **Skewness**: -0.150 (close to 0, indicating good symmetry).
   - **Kurtosis**: -0.904 (slightly below normal, indicating slightly flatter tails).
   - **Conclusion**: Likely normal.
3. **ln_overspeed**:
   - **Skewness**: -0.543 (slightly negative, just outside the symmetric range of -0.5 to 0.5, but still close).
   - **Kurtosis**: -0.718 (slightly below normal, indicating slightly flatter tails).
   - **Conclusion**: Approximate normality.





Regression Model Analysis:

1. **Model Strength (R and R^2):**

- **R (Correlation Coefficient):** 0.792
  - Indicates a strong positive relationship between the predictors (ln_alcohol and ln_overspeeding) and the dependent variable (ln_total).
- **R^2 (Coefficient of Determination):** 0.628
  - 62.8% of the variance in the dependent variable (ln_total) is explained by the predictors.
- **Adjusted R^2:** 0.625
  - Adjusted for the number of predictors, showing a slight reduction in explanatory power, which suggests that the model is well-fitted with the given predictors.

2. **Standard Error of the Estimate:**
   - Value: 1.41904
   - Indicates the average deviation of observed values from the predicted values. Lower values signify better model precision.
3. **Durbin-Watson Statistic:**
   - Value: 1.143
   - This measures autocorrelation in residuals. A value close to 2 is ideal, indicating no autocorrelation. A value of 1.143 suggests some positive autocorrelation, which might require further diagnostics.
4. **Predictors:**
   - The model uses logarithmic transformations of alcohol consumption (ln_alcohol) and overspeeding incidents (ln_overspeeding) as predictors for ln_total (the dependent variable).
   - These transformations may help linearize relationships and normalize data.
5. **Implications:**
   - Alcohol consumption and overspeeding are strong predictors of the outcome (ln_total), but residual diagnostics should be reviewed to address potential issues like autocorrelation.
   - Additional predictors or data transformations may be considered to further improve the model fit and Durbin-Watson statistic.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.792[a]	.628	.625	1.41904	1.143

a. Predictors: (Constant), ln_alcohal, ln_overspeeding

b. Dependent Variable: ln_total

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	899.603	2	449.802	292.389	.000[b]
	Residual	419.974	273	1.538		
	Total	1319.577	275			

a. Dependent Variable: ln_total

b. Predictors: (Constant), ln_overspeed, ln_alcohal

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	2.004	.297		6.743	.000		
	ln_overspeeding	.628	.063	.557	10.031	.000	.533	1.875
	ln_alcohal	.324	.060	.299	5.390	.000	.533	1.875

a. Dependent Variable: ln_total

4.4 Panel Least Squares Method

Panel Least Squares Regression Analysis:

1. **Dependent Variable:**
   - **LN_TOTAL:** The natural logarithm of the total variable is modeled as the dependent variable.
2. **Independent Variables:**
   - **LN_ALCOHAL:** Log-transformed alcohol-related data.
   - **LN_OVERSPEED:** Log-transformed overspeeding data.
3. **Regression Coefficients and Interpretations:**
   - **Intercept (C):** 0.683421
     - Represents the expected value of LN_TOTAL when LN_ALCOHAL and LN_OVERSPEED are zero.
   - **LN_ALCOHAL Coefficient:** 0.331067
     - A 1% increase in LN_ALCOHAL is associated with approximately a 0.33% increase in LN_TOTAL, holding LN_OVERSPEED constant.
   - **LN_OVERSPEED Coefficient:** 0.488470
     - A 1% increase in LN_OVERSPEED is associated with approximately a 0.49% increase in LN_TOTAL, holding LN_ALCOHAL constant.
4. **Statistical Significance (p-values):**

- All predictors (C, LN_ALCOHAL, and LN_OVERSPEED) are statistically significant at the 1% level (p-values < 0.01).
- This suggests that the predictors have a meaningful impact on the dependent variable.

5. **Model Fit and Diagnostics:**
   - **R-squared:** 0.681736
     - Approximately 68.2% of the variance in LN_TOTAL is explained by the predictors.
   - **Adjusted R-squared:** 0.679404
     - Accounts for the number of predictors, indicating a well-fitted model.
   - **Root Mean Square Error (RMSE):** 1.233549
     - Represents the average deviation of predicted values from observed values in log-transformed units.
   - **Durbin-Watson Statistic:** 1.120883
     - Indicates potential positive autocorrelation, which may need further investigation or correction.
6. **Overall Model Significance:**
   - **F-statistic:** 292.3893
     - The high value and p-value of 0.000000 confirm that the overall regression model is statistically significant.
7. **Model Selection Criteria:**
   - **Akaike Information Criterion (AIC):** 3.279408
   - **Schwarz Criterion (SC):** 3.318760
     - These criteria provide metrics for comparing models, where lower values indicate better model fit.

Key Insights:

- Both LN_ALCOHAL and LN_OVERSPEED significantly contribute to explaining variations in LN_TOTAL.
- The model explains a substantial proportion of the variance ($R^2 = 68.2\%$), but residuals may need further examination due to the low Durbin-Watson statistic.

Equation: UNTITLED Workfile: ETSA::Etsa\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LN_TOTAL
Method: Panel Least Squares
Date: 11/17/24 Time: 17:55
Sample: 2008 2015
Periods included: 8
Cross-sections included: 35
Total panel (unbalanced) observations: 276

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.683421	0.223474	3.058173	0.0024
LN_ALCOHAL	0.331067	0.048968	6.760919	0.0000
LN_OVERSPEED	0.488470	0.042640	11.45580	0.0000

Root MSE	1.233549	R-squared	0.681736
Mean dependent var	5.702101	Adjusted R-squared	0.679404
S.D. dependent var	2.190539	S.E. of regression	1.240309
Akaike info criterion	3.279408	Sum squared resid	419.9738
Schwarz criterion	3.318760	Log likelihood	-449.5583
Hannan-Quinn criter.	3.295199	F-statistic	292.3893
Durbin-Watson stat	1.120883	Prob(F-statistic)	0.000000

- **Null Hypothesis:** There are no random effects (the model does not require random effects and can be estimated using pooled OLS)
- **Alternative Hypothesis:** Random effects are present

Equation: UNTITLED Workfile: ETSA::Etsa\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Lagrange Multiplier Tests for Random Effects
Null hypotheses: No effects
Alternative hypotheses: Two-sided (Breusch-Pagan) and one-sided (all others) alternatives

	Test Hypothesis		
	Cross-section	Time	Both
Breusch-Pagan	87.12111 (0.0000)	0.953007 (0.3290)	88.07412 (0.0000)
Honda	9.333869 (0.0000)	-0.976221 (0.8355)	5.909750 (0.0000)
King-Wu	9.333869 (0.0000)	-0.976221 (0.8355)	2.969498 (0.0015)
Standardized Honda	9.866905 (0.0000)	-0.781774 (0.7828)	1.788045 (0.0369)
Standardized King-Wu	9.866905 (0.0000)	-0.781774 (0.7828)	-0.438302 (0.6694)
Gourieroux, et al.	--	--	87.12111 (0.0000)

Overall Interpretation:

- **Cross-sectional random effects are strongly supported** across all tests (p-values < 0.01).
- **Time-specific random effects are not significant**, as shown by consistently high p-values in the time dimension tests.
- When considering **both dimensions**, the results suggest significant overall random effects.
- Based on this analysis, a **random effects model with cross-sectional focus** may be more appropriate for the dataset.

REM:

Key Findings for REM Applicability:

1. **Cross-sectional Random Effects**:
    - All tests strongly reject the null hypothesis of no cross-sectional random effects (p-value = 0.0000).

- This indicates that a Random Effects Model is appropriate for capturing unobserved heterogeneity across cross-sections (e.g., individuals, firms, regions).

2. **Time-specific Random Effects**:
   - The tests do not find evidence of significant random effects over time (p-values > 0.05 for the time dimension).
   - This suggests that time-specific variability is minimal and does not require a random effects structure.

Equation: UNTITLED Workfile: ETSA::Etsa\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LN_TOTAL
Method: Panel EGLS (Cross-section random effects)
Date: 11/18/24 Time: 22:09
Sample: 2008 2015
Periods included: 8
Cross-sections included: 35
Total panel (unbalanced) observations: 276
Swamy and Arora estimator of component variances

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.772004	0.258578	10.72018	0.0000
LN_ALCOHAL	0.148837	0.040080	3.713518	0.0002
LN_OVERSPEED	0.313963	0.039268	7.995445	0.0000

Effects Specification

	S.D.	Rho
Cross-section random	0.655172	0.4219
Idiosyncratic random	0.766870	0.5781

Weighted Statistics

Root MSE	0.906683	R-squared	0.297259
Mean dependent var	2.187587	Adjusted R-squared	0.292111
S.D. dependent var	1.072970	S.E. of regression	0.911651
Sum squared resid	226.8923	F-statistic	57.73936
Durbin-Watson stat	1.418714	Prob(F-statistic)	0.000000

H0 – REM is preferred
H1 – FEM is preferred

Equation: UNTITLED Workfile: ETSA::Etsa\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Correlated Random Effects - Hausman Test
Equation: Untitled
Test cross-section random effects

Test Summary	Chi-Sq. Statistic	Chi-Sq. d.f.	Prob.
Cross-section random	114.988494	2	0.0000

Cross-section random effects test comparisons:

Variable	Fixed	Random	Var(Diff.)	Prob.
LN_ALCOHAL	0.047712	0.148837	0.000261	0.0000
LN_OVERSPEED	0.118875	0.313963	0.000570	0.0000

Cross-section random effects test equation:
Dependent Variable: LN_TOTAL
Method: Panel Least Squares
Date: 11/18/24 Time: 22:13
Sample: 2008 2015
Periods included: 8
Cross-sections included: 35
Total panel (unbalanced) observations: 276

Variable	Coefficient	Std. Error	t-Statistic	Prob.

As, FEM is preferred,

1. **Model Performance Metrics:**

- **R-squared = 0.894892**:
  - Indicates that approximately 89.49% of the variation in LN_TOTAL is explained by the independent variables and fixed effects.
- **Adjusted R-squared = 0.875411**:
  - Accounts for the number of predictors and indicates a strong fit for the model.
- **F-statistic = 45.93629 (p-value = 0.0000)**:
  - Overall model is highly significant, rejecting the null hypothesis that all coefficients are jointly zero.
- **Durbin-Watson stat = 1.910728**:
  - Suggests no severe autocorrelation in the residuals (values close to 2 indicate no autocorrelation).
- **Root MSE = 0.708892**:

- Measures the standard deviation of residuals, providing an estimate of the model's predictive accuracy.

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2. Fixed Effects Specification:

- **Cross-section fixed (dummy variables)**:
  - Accounts for unobserved heterogeneity across cross-sections (e.g., individuals, firms, or regions).
- **Period fixed (dummy variables)**:
  - Controls for unobserved effects that vary over time (e.g., policy changes, macroeconomic conditions).

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3. Key Takeaways:

1. **Significant Predictor**:
   - LN_OVERSPEED has a positive and significant impact on LN_TOTAL, suggesting that overspeeding is a key factor influencing the dependent variable.
2. **Non-significant Predictor**:
   - LN_ALCOHAL does not significantly affect LN_TOTAL in this model, though it may still be relevant in other contexts or with different specifications.
3. **Model Fit**:
   - The high R-squared and F-statistic suggest the model explains a large proportion of the variation in the dependent variable.
4. **Policy Implication**:
   - Strategies aimed at reducing overspeeding may help lower the dependent variable (e.g., accidents or fatalities, depending on the context).

5. Discussion and Conclusion

Model Selection:

Multiple statistical tests (Breusch-Pagan, and Hausman test) were conducted to determine the appropriate model
The **Hausman test** confirmed that **Fixed Effects Model (FEM)** is more appropriate than the Random Effects Model (REM), as unobserved factors are correlated with the explanatory variables.

Fixed Effects Insights

- FEM results suggest that controlling for unobserved cross-sectional differences (e.g., regional factors, infrastructure, or enforcement policies) improves the reliability of the model.

Implications

- **LN_ALCOHAL:** While its coefficient is positive, it is **not statistically significant** (p-value > 0.05). This suggests that alcohol-related factors may not have a robust impact on Ln_Total_Accidents
- **LN_OVERSPEED:** The variable is **statistically significant**, showing a meaningful relationship with the dependent variable. Over speeding seems to be a significant contributor to the changes in Ln_Total_Accidents
- **Model fit:** The high R-squared and Adjusted R-squared values indicate the model explains most of the variation in LN_TOTAL after accounting for fixed effects.

Recommendations

1. **Focus on policy measures targeting overspreading.**
    1. Since overspreading has a larger impact, stricter enforcement of speed limits and penalties for speeding violations should be prioritized.
2. **Address unobserved factors.**
    1. Variables such as road conditions, urbanization, or law enforcement intensity could further explain accident rates and should be explored in future studies.

Limitations

- **Misses constant factors:** The model ignores factors that don't change over time (like geography or culture), even if they matter.
- **Cause-and-effect issues:** It's hard to tell if the independent variables (like alcohol or over speeding) are causing the changes, or if something else is influencing both.
- **Uneven data:** Not all entities have data for every year, which can affect the accuracy of the results.

6. References

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